



Choosing a CPU

Market Overview

Model Ranges

CPU Specs

Socket Types

Model Numbers

Does it come with a cooler?

Overclockability



The Intel logo is displayed in blue lowercase letters with a registered trademark symbol.

Desktop CPU Ranges

The AMD logo is displayed in white uppercase letters on a black background, with a stylized triangle symbol.

Core
i3

Ryzen
3

Budget
~ \$100-\$150

Core
i5

Ryzen
5

Midrange
~ \$150-\$300

Core
i7

Ryzen
7

High-End
~ \$300-\$450

Core
i9

Ryzen
9

Enthusiast
~ \$450-\$700+

Thread
ripper
PRO

Extreme
~ \$2000-\$7000+

CPU Specs

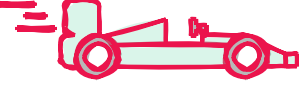
GHz



1 GHz = 1,000,000,000 / per second

CPU's have a base clock and a boost clock

Headline figure is usually the highest (boost) speed

10 years ago 3.4 GHz was good 

Progression for raw speed has slowed....

Cache - L1, L2, L3

L1 = Fastest

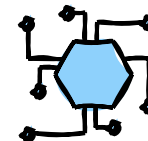
L2 = Faster than L3

L3 = Faster than RAM

RAM = Pretty fast



Socket



TDP/Power

Thermal Design Power

Maximum power consumed at highest speeds

Coolers can list TDP they will cool



Intel Core i7-12700K Gaming Desktop Processor with Integrated Graphics and 12 (8P+4E) Cores up to 5.0 GHz Unlocked LGA1700 600 Series Chipset 125W

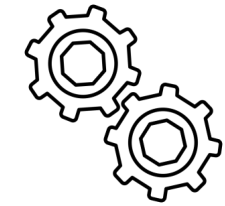
[Visit the Intel Store](#)

4.9 ★★★★★ 2,474 ratings | [Search this page](#)

- 25 MB of L3 cache to boost the instruction processing and system performance
- Integrated Intel UHD 770 Graphics
- The processor features Socket LGA-1700 socket for installation on the PCB

Single P-core Max/Base Speed	5.0/3.6 GHz
Single E-core Max/Base Speed	3.8/2.7 GHz

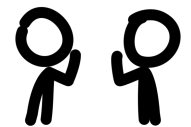
Cores / Threads



Core = Physical hardware core

(Performance or Efficient)

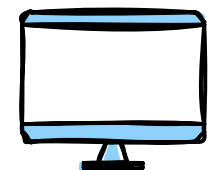
Thread = Virtual execution separation

How many cores? 4, 6 or 8 

More is not always better...

Software needs to be designed to utilize them...

Graphics





Desktop CPU Generations



2023

AMD Ryzen 7000 (Raphael)
Intel Core 14th Gen (Meteor Lake)

6 GHz

2022

Intel Core 13th Gen (Raptor Lake)

6 GHz

2021

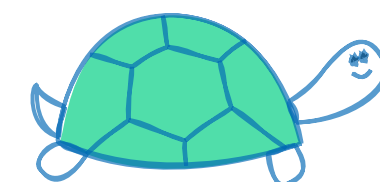
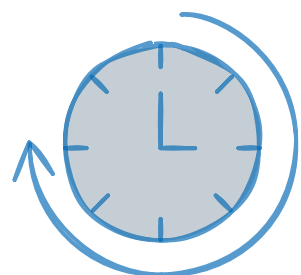
Intel Core 12th Gen (Alder Lake)
Intel Core 11th Gen (Rocket Lake)

5.2 GHz

2020

AMD Ryzen 5000 (Vermeer)

4.8 GHz





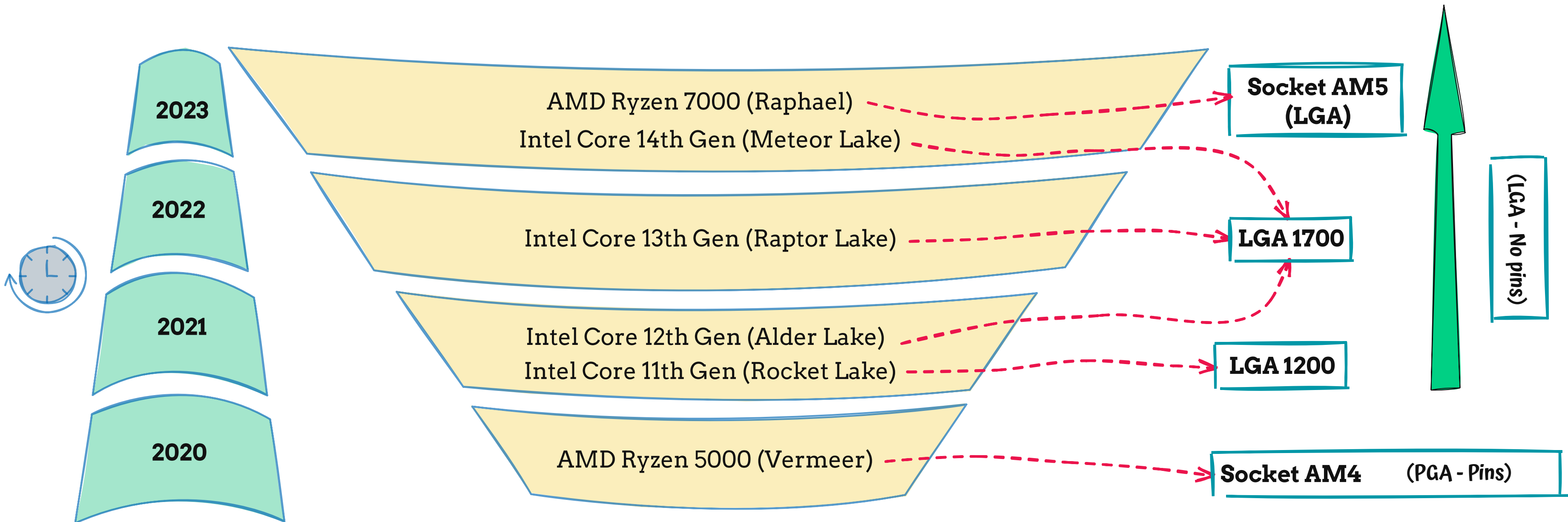
PGA = Pin Grid Array



Desktop CPU Sockets



LGA = Land Grid Array



So it's an i5 (but which series? how old?)



Intel Core i5-12600k 4.9 GHz (Alder Lake)



Intel Core i5-13600k 5.1 GHz (Raptor Lake)

Both use socket LGA 1700

So it's an Ryzen 5 (but which series? how old? what socket?)



AMD Ryzen 5 5600X 4.6 GHz (Socket AM4)

AM4 is a PGA socket (has pins)



AMD Ryzen 5 7600X 5.3 GHz (Socket AM5)

AM5 is an LGA socket (no pins)

Desktop CPU Model Numbers

AMD

AMD Ryzen 5 7600X

Brand

Range

Generation

SKU

Suffix

X = High Performance
X3D = 3d V-Cache tech (better for gaming)
G = On board graphics (only used in ranges where graphics are optional)
T = Designed for power efficiency (less performance, less power usage) (not used in latest models)

intel

Intel Core i7-13700K

Brand

Range

Generation

SKU

Suffix

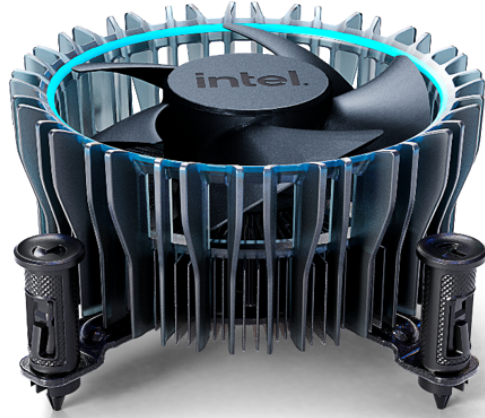
13th Gen (Raptor Lake)

No suffix = Locked, with on-board graphics
K = unlocked (overclockable)
F = no on-board graphics
KF = unlocked and no on-board graphics
T = designed for power efficiency (less performance, less power usage)
X = Xtreme edition
S = special edition

It's not always obvious, or clear which model does what.... always lookup the exact model on the AMD or Intel website....

intel®

Intel Laminar



Does it come
with a cooler?

AMD

Wraith Prism



All Intel® Desktop Processors that have these letters in
their number

K, KF, KS, XE, X

Do not come with a cooler....

Examples:

i7-14700K, i5-13600KF, i9-12900K, i7-11700K,
i9-1200KF, i9-12900KS, i9-9980XE, and i5-10600K

Do you know why?

In general efficient or lower power CPUs (Low TDP) come
with a cooler

The AMD Website CPU specification pages list whether
a Thermal Solution is included

AMD Ryzen™ 9 7900	
Regional Availability: Global, China, NA, EMEA, APJ, LATAM	Platform: Desktop
Product Line: AMD Ryzen™ 9 Processors	# of CPU Cores: 12
Max. Boost Clock: Up to 5.4GHz	Base Clock: 3.7GHz
L2 Cache: 12MB	L3 Cache: 64MB
Processor Technology for CPU Cores: TSMC 5nm FinFET	Unlocked for Overclocking
Thermal Solution (PIB): AMD Wraith Prism	Max. Operating Temperature: 95°C
*OS Support: Windows 11 - 64-Bit Edition Windows 10 - 64-Bit Edition RHEL x86 64-Bit Ubuntu x86 64-Bit *Operating System (OS) support will vary by manufacturer.	



Can I overclock
it?



If it has a K or an X
in the suffix

YES!

Well except for the X3D gaming chips....

Examples:

i7-14700K, i5-13600KF, i9-12900K, i7-11700K,
i9-1200KF, i9-12900KS, i9-9980XE, and i5-10600K

Check the manufacturer website....